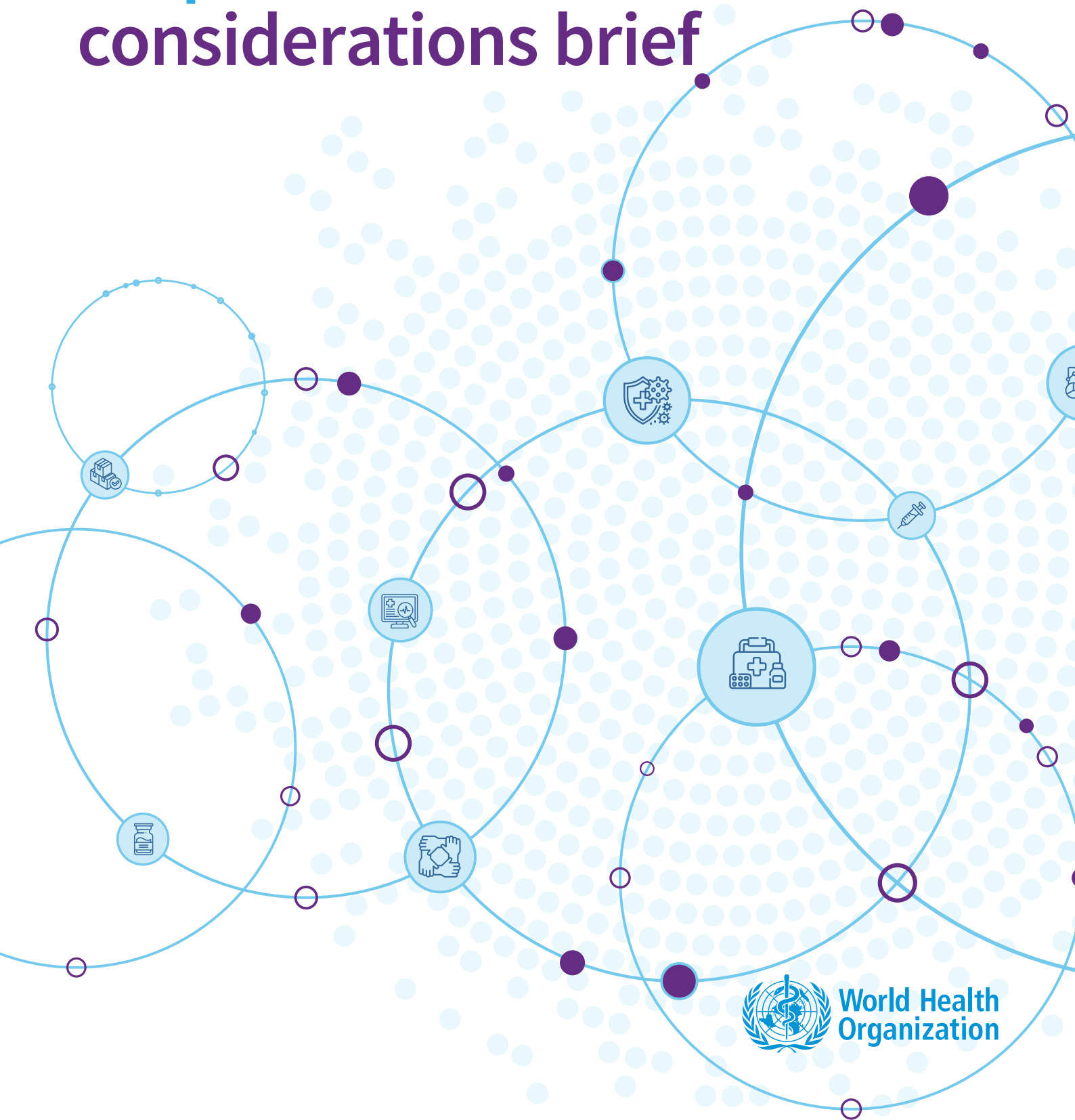

Stockpiling of medical countermeasures for pandemic influenza considerations brief



World Health
Organization

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ISBN 978-92-4-011786-0 (electronic version)

ISBN 978-92-4-011787-7 (print version)

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Cataloguing-in-Publication (CIP) data. CIP data are available at <https://iris.who.int/>.

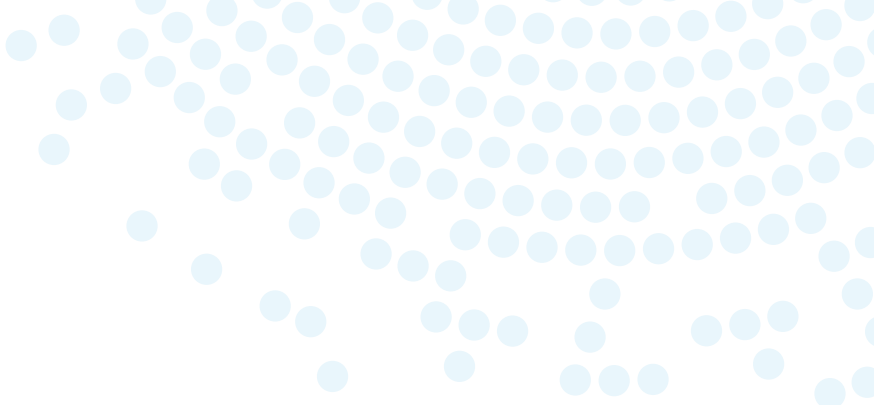
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Acknowledgements

The World Health Organization (WHO) expresses its gratitude to the many individuals involved in the development of this document.

The development of the document was coordinated by Ioana Ghiga, who also developed the technical concept, study design and supervised data collection, analysis, writing or review (Epidemic and Pandemic Management Department, Medical Countermeasures Unit). Marie Ryan who worked as a consultant in the Medical Countermeasures Unit, undertook evidence review or synthesis, data collection, analysis and writing.

The document was circulated to global and regional colleagues for review and comments; where input was provided (names are mentioned), it informed the content of this considerations document:

- WHO headquarters units or teams: Medical Countermeasures (Tim Nguyen, Shoshana Goldin, Oluwatosin Akande), Global Respiratory Threats (Sarah Hess), Expanded Programme on Immunization; Regulatory Systems Strengthening, Operation Support and Logistics (Paul Molinaro), Clinical Management and Operations.
- WHO regional offices representatives: WHO Regional Office for Africa; WHO Regional Office for the Americas (Francisco Nogareda); WHO Regional Office for South-East Asia; WHO Regional Office for Europe; WHO Regional Office for the Eastern Mediterranean; and WHO Regional Office for the Western Pacific.

The document was formulated in accordance with the work plan under the implementation of the PIP Framework. WHO expresses its gratitude to the PIP Frameworks Partnership Contributions for supporting the development and publication of this document. No external financial contributions were received for this work.

Abbreviations

DALY	Disability-adjusted life year
KPI	Key performance indicators
IPC	Infection prevention and control
MCM	Medical countermeasures
NRA	National Regulatory Authority
PA	Procurement Agent
PPE	Personal protective equipment
QALY	Quality-adjusted life year
QC	Quality Control
WHO	World Health Organization

Executive summary

Influenza pandemics have historically imposed profound and multifaceted public health, societal, and economic burdens, underscoring the imperative for coordinated, multisectoral preparedness strategies. Central to these strategies is the equitable and timely access to medical countermeasures (MCM)—encompassing vaccines, therapeutics, diagnostics, medical devices, and ancillary supplies—which are critical components for mitigating the impact of pandemic influenza as well as related respiratory threats. This document presents a comprehensive, risk-informed framework alongside a suite of considerations designed to assist countries in evaluating the potential role, strategic design, and operational management of MCM stockpiles within their overarching pandemic preparedness and response architectures.

The document situates stockpiling as one of several complementary mechanisms intended to enhance MCM availability during public health emergencies, explicitly distinguishing it from routine procurement, supply chain management, or health system logistics. Stockpiling modalities vary widely, ranging from traditional physical reserves to virtual or contract-based access arrangements, each presenting distinct operational, financial, and logistical considerations and trade-offs. Acknowledging the inherent complexity and substantial resource commitments required, countries are encouraged to adopt a methodical, risk-based approach—carefully balancing assessments of hazard likelihood and severity, system vulnerabilities, and national capacities—to inform stockpiling decisions.

Key risk dimensions warranting comprehensive assessment include:

- **Hazard likelihood and severity:** This dimension extends beyond the threat of pandemic influenza alone to encompass the cumulative burden of seasonal influenza, zoonotic spillovers, and the circulation of other respiratory pathogens, recognizing that recurrent outbreaks may contribute to the depletion or strain of existing stockpiles, thereby influencing replenishment strategies and surge capacity planning.
- **Vulnerabilities:** These span multiple domains, including health system constraints such as limited domestic manufacturing capabilities, dependence on global supply chains, fragility in procurement and distribution networks, and workforce limitations. Economic vulnerabilities pertain to the availability and sustainability of financial resources dedicated to MCM procurement, maintenance, and contingency scale-up. Social vulnerabilities incorporate issues of equity, humanitarian considerations, and sociopolitical instability, all of which may exacerbate pandemic consequences and complicate effective response.
- **National capacity:** A nuanced understanding of a country's financial and human resources, physical infrastructure, and regulatory environment is critical to evaluating the feasibility of establishing and sustaining a stockpile over time. Complementing this are existing legal frameworks, policy environments, and institutional mechanisms that underpin effective stockpile governance, quality assurance, and emergency deployment capabilities.

Decisions regarding the establishment and maintenance of stockpiles should be firmly grounded in these multidimensional assessments and contextualized within the country's epidemiological realities, economic constraints, and operational capacities. In some contexts, alternative mechanisms—such as regional stockpile pooling, mutual aid agreements, or contingent procurement arrangements—may offer more pragmatic or resource-efficient solutions than maintaining a dedicated national stockpile.

Where stockpiling is pursued, it is essential that clearly articulated goals and objectives are established. These objectives may encompass enabling rapid national response capabilities, ensuring surge capacity within healthcare facilities under stress, mitigating supply chain disruptions, supporting large-scale public health interventions including vaccination campaigns, enhancing national security preparedness, and promoting cost-effectiveness through strategic resource allocation.

Planning for stockpile composition and management entails deliberate decisions on the selection of MCM types and quantities, procurement strategies, storage infrastructure, maintenance protocols, and continuous updating to incorporate advances in medical technology and shifts in pathogen characteristics. Forecasting methodologies benefit from integrating historical epidemiological data, proxy indicators, and scenario-based modeling to optimize preparedness thresholds. Procurement processes must prioritize transparency, adherence to stringent quality standards, fiscal prudence, and where appropriate, leverage collaborative purchasing mechanisms or innovative contractual models to enhance affordability and supply stability.

Robust policy frameworks and governance structures are foundational to effective stockpile stewardship. This necessitates explicit delineation of roles, responsibilities, operational procedures, and mechanisms for stakeholder engagement across government sectors, private partners, and international collaborators. Compliance with pertinent legal and regulatory requirements—including customs protocols, environmental regulations, intellectual property rights, and emergency-specific liability provisions—is essential. Coordination with National Regulatory Authorities (NRAs) is essential to ensuring product quality, safety, and facilitating expedited authorization and recall processes when necessary.

Supply chain resilience constitutes an important element for ensuring stockpile efficacy. This entails diversification of sourcing channels, seamless integration of internal logistics with external manufacturing and distribution networks, leveraging digital technologies for real-time inventory tracking and traceability, and operational agility to respond to disruptions. Storage facilities must conform to established best practice standards and be supported by robust, secure inventory management systems. Equitable distribution strategies should proactively address access challenges faced by vulnerable, marginalized, and geographically isolated populations, necessitating coordination among various stakeholders.

A multidisciplinary, highly skilled workforce proficient in procurement, inventory control, logistics management, and emergency deployment is desirable. Ongoing capacity building through continuous professional development, simulation exercises, and effective risk communication enhances operational readiness and mitigates response risks during crisis events.

Governance arrangements typically involve a central coordinating entity empowered with multisectoral representation, responsible for overseeing resource allocation, policy enforcement, operational management, and stakeholder coordination. Performance monitoring and evaluation systems employing key performance indicators—such as stock accuracy, timeliness of access, quality compliance, and cost-efficiency—facilitate data-driven continuous improvement. Periodic reviews and joint cross-border simulation exercises further reinforce preparedness and operational coherence across jurisdictions.

This document aims to offer a flexible framework to support countries in making contextually appropriate, risk-calibrated decisions concerning MCM stockpiling for pandemic influenza. By judiciously balancing hazard assessments, systemic vulnerabilities, national capacities, and resource constraints, countries can strengthen their pandemic preparedness and resilience, optimizing the strategic availability and deployment of critical MCM amid the evolving landscape of infectious disease threats and operational complexities.

1. Introduction

Influenza pandemics throughout history have had significant and long-lasting impacts on public health, society, and the economy. To mitigate these consequences, countries must plan for a coordinated, multisectoral response over the long term. A key aspect of pandemic preparedness is access to medical countermeasures (MCM) including vaccines, medicines, medical devices and equipment. In the early stages of a pandemic, some existing MCM, such as antivirals, may be repurposed based on emerging efficiency data. Additionally, pandemics with high morbidity and mortality rates can overwhelm healthcare systems, leading to severe shortages of medical supplies (1).

It is important to recognize that MCM should be considered in conjunction with public health and social measures and other interventions, as part of a comprehensive response strategy. The WHO guidance on pandemic preparedness emphasizes the integration of MCM deployment with public health and social measures to maximize the effectiveness of both approaches in controlling disease transmission and mitigating impact (2). WHO has also developed the WHO guideline on public health and social measures for mitigating the risk and impact of epidemic and pandemic influenza, which provides detailed recommendations on public health and social measures implementation during influenza pandemics (3).

Pandemic planning must consider the potential for shortages and disruptions in the availability of MCM. This necessity is emphasized in various interpandemic planning recommendations, including the WHO guidance on pandemic preparedness for respiratory viruses (2). WHO advises countries to evaluate the necessity of stockpiling specific MCM and, if deemed necessary, to enhance their stockpiling capabilities accordingly (2,4)¹.

For this document, the term stockpiling will be used broadly to encompass a number of different approaches to assure access to critical MCM in a clinically relevant timeframe. These range from traditional physical stockpiles where the required MCM is held under direct control in physical warehouses available for immediate deployment to contracted access to MCM (virtual stockpiles) in specified quantities and timeframes. Each approach has different strengths and cost implications. Regardless of the approach, effective stockpiling involves not only the ability to ensure access to required MCM but also the strategic management of these resources (e.g. including through periodic replenishment).

1.1 Purpose and scope of this document

This document aims to support countries in making strategic and operational decisions by providing a framework to: i) assess the need for stockpiling; and ii) guide the planning, establishment, and management of stockpiles if a decision to stockpile is made. The scope of this document specifically addresses MCM for pandemic influenza, as outlined in the WHO Disease Commodity Package (5), with a primary focus on products intended for human use. In addition, WHO has published clinical practice guidelines for influenza management which provide evidence-based recommendations to guide healthcare providers on the appropriate use of MCM during seasonal and pandemic influenza (6).

1 WHO Health Emergency Prevention, Preparedness, Response and Resilience meant to support strengthening of global and national capacities for effectively preparing for and responding to health emergencies. It advises strengthening capabilities across five sub-systems: (1) *collaborative surveillance*, (2) *community protection*, (3) *safe and scalable care*, (4) *access to countermeasures*, and (5) *emergency coordination*. Capabilities related to stockpiling are mentioned under safe and scalable care and access to countermeasures (*please see Annex 1*).

2. Decision-making considerations for establishing stockpiles

To underpin decision-making on whether to establish and maintain stockpiles, a risk assessment approach has been adopted, drawing conceptually on the Strategic Toolkit for Assessing Risks (7). This framework is premised on the recognition that risk analysis provides an evidence-based method to evaluate hazards, vulnerabilities, and national capacities, thereby ensuring that resource allocation is aligned with actual health system needs and broader societal priorities. These domains are directly adapted from the Strategic Toolkit for Assessing Risks' three core areas of analysis—hazards, likelihood, and impact—while reframing them to capture the specific challenges of MCM stockpiling. In particular, the Strategic Toolkit for Assessing Risks emphasis on hazard identification (scale, exposure, health consequences), likelihood assessment (frequency, seasonality, probability), and impact (severity, vulnerability, coping capacity) has been translated into stockpiling-relevant considerations, such as competing pressures on limited reserves, the equity of access pathways, and the durability of national investment in preparedness infrastructure.

While it may not be possible to know with certainty the efficacy of all MCM in advance of a pandemic, an essential consideration in stockpiling decisions is the extent to which available evidence supports the likely effectiveness and safety of these products. Decisions should be informed by the best available data. Stockpiling should not rely solely on supply-side readiness but also on a reasonable expectation of clinical utility across different population groups and use-case scenarios. Where uncertainty exists, mechanisms for rapid evidence generation, post-deployment evaluation, and adaptive guidance should be integral to the stockpiling strategy.

The risk analysis would imply understanding the likelihood and severity of a hazard posed by a novel influenza virus with pandemic potential, and how this relates to more frequent national risks stemming from seasonal influenza, zoonotic influenza or other respiratory viruses. These more predictable outbreaks might call on the same reserves of MCM that may be used for a pandemic response, either from the health system or from routine stockpiles, causing fewer products to be available if they coincide with a pandemic event.

A second risk dimension examines the vulnerabilities according to the choice of access and the potential consequences for the health system, society, and the economy.

A third risk dimension looks at the resource commitment needed from the government for stockpiling, for both the initial set-up and long-term maintenance. This risk dimension (national capacity) is gauged by the investment assigned for infrastructure, human and financial resources and (if needed) for updating policies and regulations.

The framework includes broad questions for each risk component to determine if MCM stockpiling is appropriate given the country's context.

A decision not to establish stockpiles for MCM will be because the country will have determined from the risk assessment that existing measures will be adequate for rapid access to MCM in sufficient quantities to respond to a pandemic crisis. Such a decision may be due to other access arrangements such as regional, pooled or other agreements, or because the country cannot commit any resources for pandemic preparedness.

Stockpiles can have different goals and objectives. These may evolve over time and therefore the current framework considers the continuous feedback loop between establishing, maintaining and re-evaluation of stockpile characteristics as necessary.

The overall stockpiling framework is presented in Figure 1., while more detailed considerations to aid decision to set-up or not set-up are illustrated in Figure 2.

Figure 1. MCM stockpiling considerations framework

Risk Profile

considerations on whether to establish /or not establish MCM stockpile(s)

Described according to three risk categories, related risk sub-categories & exploratory questions

Goals & Objectives

may evolve because of changes in the Risk Profile therefore periodic re-evaluation of risks is necessary

Planning considerations

for the set-up and management of MCM stockpile(s)

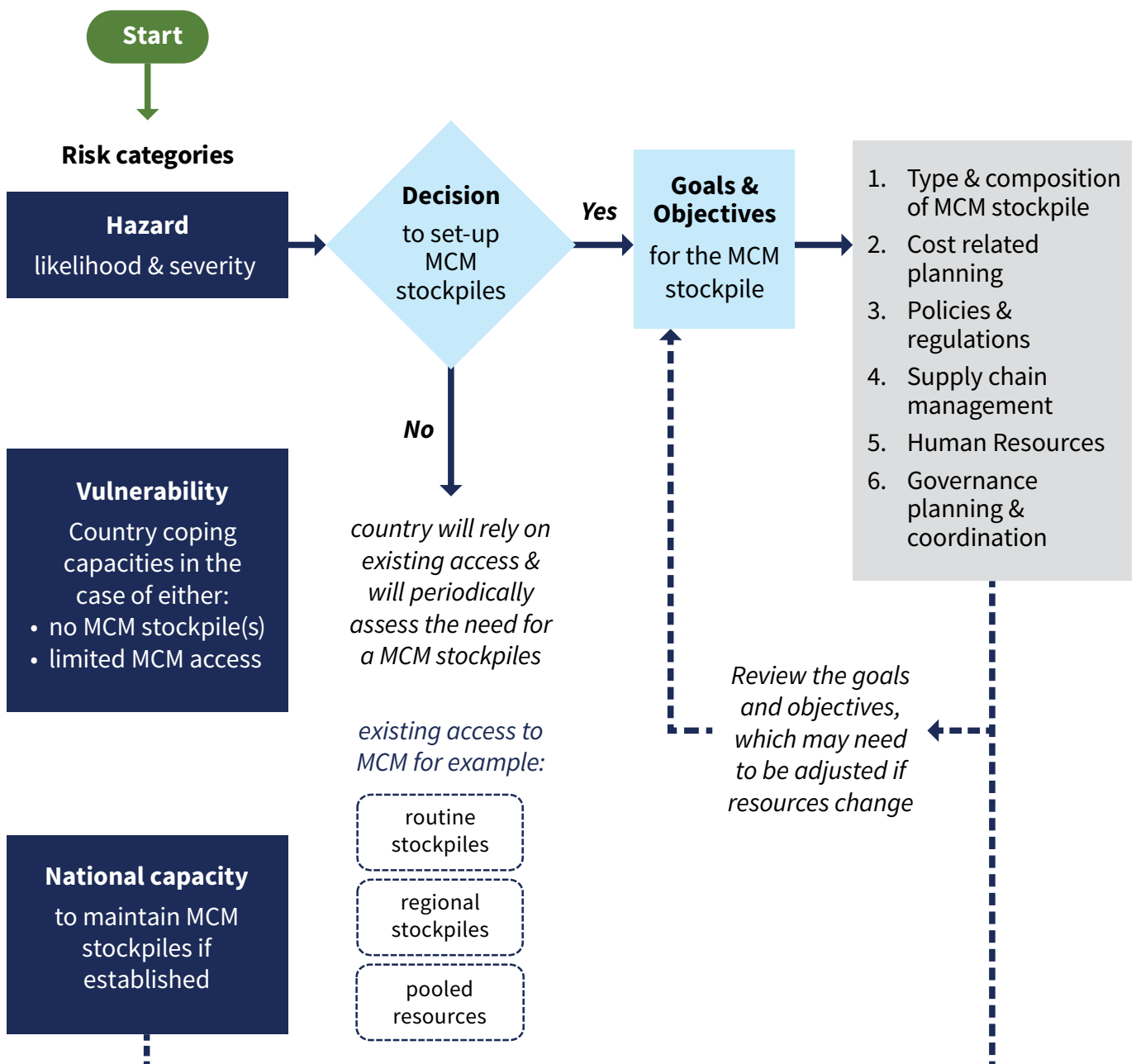
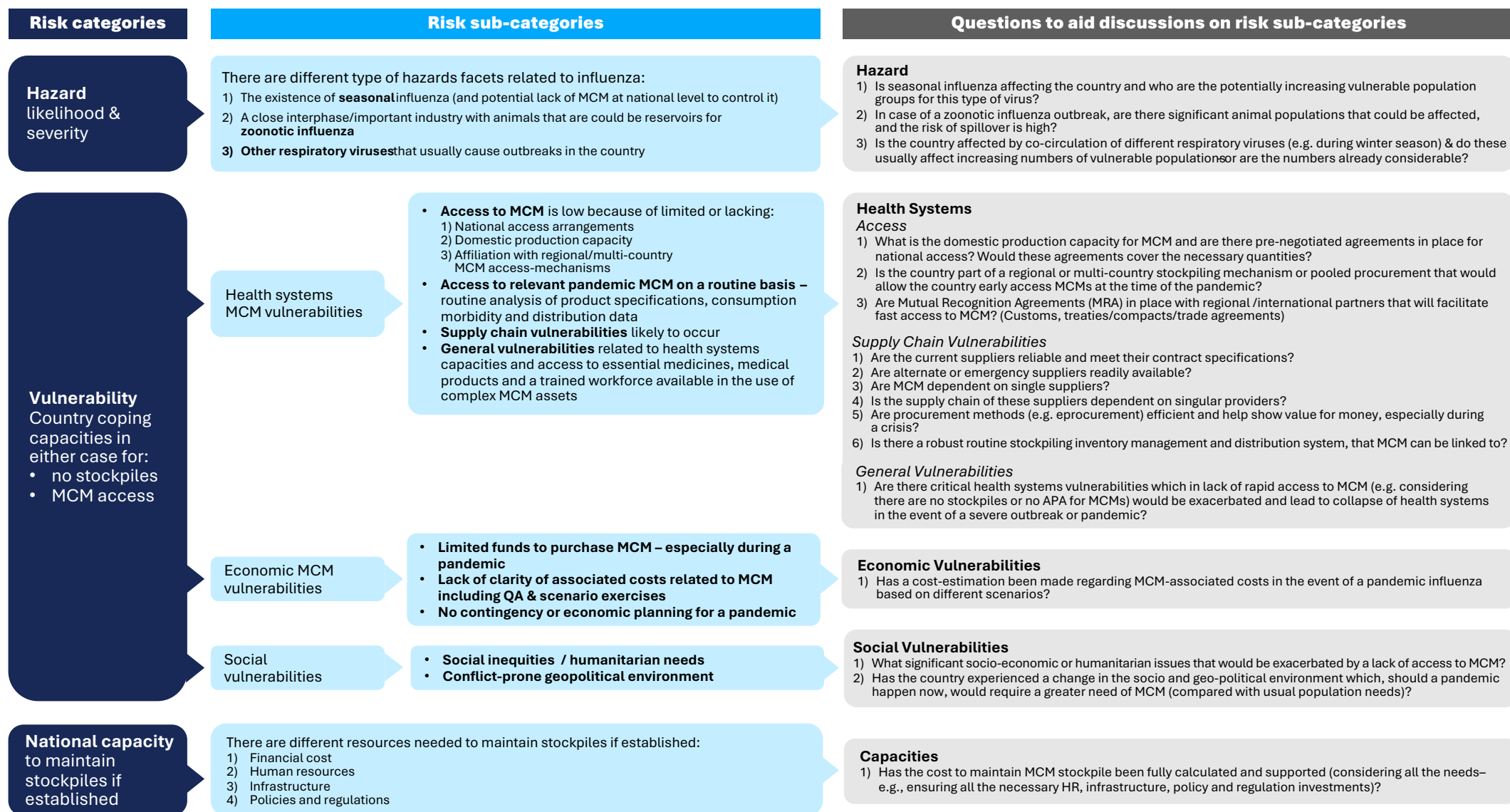


Figure 2. Detailed framework for assessing the risks in view of deciding to establish /not establish national MCM stockpiles



2.1 Hazard likelihood and severity

When evaluating stockpiling for pandemic influenza MCM, countries should consider a wider range of related risks, such as seasonal influenza, zoonotic influenza, and other respiratory viruses. This approach can offer potential benefits in two main ways:

1. It provides an opportunity to effectively manage resources to address recurrent and predictable disease risks.
2. If there is some level of stockpiling in place, it might be helpful to know how predictable disease outbreaks can rapidly deplete MCM stocks, and what would be the likely outcome if any of these disease risks were to coincide with a pandemic event.

Authorities should assess if their current access to MCM is adequate for managing seasonal influenza and protecting vulnerable populations. Similarly, determining the risk of zoonotic influenza is important especially for locations where there is a high degree of captive populations or significant interaction between humans and animals, or a history of zoonotic influenza cases. Finally, countries should examine what other co-circulating respiratory viruses are likely to elevate the risk, increase the burden on the healthcare system, and potentially creating larger vulnerable populations during a pandemic emergency.

2.2 Vulnerability and country coping capacities

The types of vulnerabilities that make it difficult for a country to respond to a pandemic without MCM stockpiling arrangements or with limited access to MCM are described from three different perspectives: *health systems*, *economic*, and *social vulnerabilities* (Figure 2).

2.2.1 Health systems vulnerabilities

In assessing health systems vulnerabilities in relation to pandemic influenza MCM stockpiling, countries may find it beneficial to analyze several potential vulnerabilities in access mechanisms for MCM, supply chains or more general within the health system.

One consideration is to evaluate **domestic access to MCM**, particularly if access is limited or lacking due to factors such as:

- Limited or untimely national access arrangements. Questions to be considered would be looking at whether there are established national agreements or policies in place that ensure timely access to MCM during a pandemic (including considerations on whether MCM may be part of the national essential medicines list)?
- Limited or insufficient domestic production capacity. Decisionmakers should consider whether the country has the capacity to produce necessary MCM domestically, or is it reliant on external sources? How reliable is MCM production sourced from external suppliers during an emergency? What is the domestic production capacity for MCM, and whether there are pre-negotiated agreements for national access? Would these agreements cover the necessary quantities during a pandemic?
- Limited or no affiliation with regional or multi-country MCM access mechanisms. A question to be explored is whether the country is part of any regional or international stockpiling or procurement mechanisms that could facilitate access to MCM during a crisis?

Existing vulnerabilities within the **supply chain** may also increase the risk in accessing MCM at the time of the pandemic, and therefore may increase the need for stockpiling. Potential supply chain vulnerabilities can be summarized as disruptions due to limited sourcing options, and ability of suppliers to respond to extreme demands in short timeframes, considering potential disruptions due to the pandemic.

Countries may want to assess whether current suppliers are dependable and whether they meet contract specifications consistently. Additional questions to consider include:

- Are suppliers reliable and meet contract specifications?
- Are alternate or emergency suppliers readily available in case of disruptions?
- *Are MCM dependent on single suppliers, and is the supply chain for these suppliers reliant on singular providers?* This analysis might extend to examining procurement methods, such as whether e-procurement systems are in place and whether they contribute to efficiency and value for money, especially during a crisis.
- Is there a robust routine stockpiling inventory management and distribution system that can be effectively linked to MCM?

These considerations can aid analysis whether internal supply chains remain intact during a pandemic, avoiding potential bottlenecks that could impede access to essential MCM.

Lastly, it could be useful to explore **general vulnerabilities** within the health system that might be exacerbated by the lack of stockpiles during a pandemic influenza. For instance, countries might ask: *Are there critical vulnerabilities within the health system that, in the absence of rapid access to MCM (e.g., due to a lack of stockpiles or advance purchase agreements), could lead to the collapse of health services during a severe outbreak or pandemic?* This could involve assessing the availability of essential medicines, medical products, and the presence of a trained workforce capable of deploying complex MCM assets effectively. Identifying and addressing these vulnerabilities can be crucial in ensuring that health systems remain resilient and capable of responding to the demands of a pandemic.

2.2.2 Economic vulnerabilities

When considering the establishment of stockpiles for pandemic influenza MCM, it is important for countries to evaluate potential economic vulnerabilities that could exacerbate risks if stockpiles are not in place. In resource-constrained settings, such vulnerabilities may be compounded by structural limitations, including narrow fiscal space, reliance on external suppliers, or dependence on donor support. During global crises—such as the COVID-19 pandemic—many low- and middle-income countries faced difficulties securing MCM in a competitive international market, where high-income countries were able to enter into advance purchase agreements at elevated prices. However, these challenges are not uniform, and countries may vary in their exposure to such risks depending on procurement capacity, regulatory agility, and existing regional or global partnerships. These considerations highlight the importance of undertaking context-specific cost-benefit analyses (see Annex 2) and exploring collaborative, cost-effective alternatives—including regional pooled procurement or shared stockpiling mechanisms—which can help distribute risk and reduce the resource burden on individual countries.

In assessing this risk domain, one key consideration is the availability of funds to purchase MCM, particularly in the midst of a pandemic when demand and prices are likely to surge. Countries may want to assess whether they have adequate financial reserves or contingency funds specifically allocated for the procurement of MCM in emergency situations. Questions to consider include: *Does the country have sufficient funds set aside to purchase MCM during a pandemic, when prices and demand are likely to be high?*

Another area to explore is the clarity and understanding of the full range of costs associated with MCM, including not just the purchase price, but also costs related to quality assurance, storage, maintenance, and scenario exercises. *Has the country conducted a comprehensive cost estimation of MCM-associated expenses in the event of a pandemic influenza?* This might involve analyzing different scenarios to understand the financial implications under various conditions, ensuring that the country is prepared for the economic demands of a pandemic.

Additionally, the absence of contingency or economic planning for pandemics can leave a country vulnerable. Countries might ask: *Is there a clear economic plan or contingency strategy in place for financing MCM during a pandemic?* This could include exploring options for accessing emergency funds, engaging in pre-negotiated purchasing agreements, or participating in international financial mechanisms designed to support pandemic response efforts.

2.2.3 Social vulnerabilities

When planning for the establishment of stockpiles for pandemic influenza MCM, countries should also consider social vulnerabilities that could exacerbate risks if these stockpiles are not in place. These vulnerabilities are particularly significant in contexts marked by social inequities, humanitarian needs, and conflict-prone geopolitical environments.

One of the primary considerations is the potential for deepening social inequities and existing humanitarian needs because of lack of timely access to MCM. While in some socio-economic strata certain public health and social measures (e.g. safely staying at home) may be possible, in others this may not be option. Therefore, a key question to consider is: *What significant socio-economic or humanitarian issues could be exacerbated by a lack of access to MCM?*

Another critical aspect to consider is the geopolitical context, particularly in regions prone to conflict or instability. Countries might assess whether their socio-political environment has changed in ways that could increase the demand for MCM during a pandemic. *Has the country experienced changes in its socio-political environment that would heighten the need for MCM if a pandemic occurred now, compared to usual population needs?* In conflict-prone or politically unstable regions, the strain on healthcare systems and the increased vulnerability of certain populations could create a greater urgency for MCM, making the availability of stockpiles even more critical.

Additionally, it may be valuable to consider how social vulnerabilities could influence public trust and cooperation in the distribution of MCM. In areas with significant social tensions or mistrust of government institutions, the absence of timely access to MCM could exacerbate social unrest and hinder pandemic response efforts.

2.3 National capacity to maintain stockpiles if established

When considering whether to establish stockpiles for pandemic influenza MCM, it is essential for countries to evaluate their national capacities to maintain these stockpiles over time. This involves assessing the various resources required, including financial costs, human resources, infrastructure, and the necessary policies and regulations. Understanding these capacities will facilitate making informed decisions about whether stockpiling is a feasible and sustainable option.

One of the primary considerations is related to **costs** including ongoing financial commitment required to maintain stockpiles. Countries should assess whether they have the budgetary resources to support not just the initial procurement of MCM, but also the long-term costs associated with storage, maintenance, and eventual replenishment of these stockpiles. A key question to consider is: *Has the full cost of maintaining an MCM stockpile been thoroughly calculated? Is the financial support available to cover these costs over time? Is the investment in stockpiling captured in law?* In estimating the full costs, countries should consider all the types of potential costs. Establishing and maintaining virtual or physical stockpiles involves significant sunk costs for personnel, infrastructure, procurement, setup, and ongoing maintenance. These costs may be incurred over many years in anticipation of a crisis event. As the goals and objectives of MCM stockpiles evolve with changing disease surveillance information, countries need to estimate these financial commitments thoroughly. Countries might explore various cost analysis methods to understand the financial implications. Simple methods such as cost-benefit analysis, detailed reviews of direct and indirect costs, or more complex economic assessments like cost-effectiveness or comparative-effectiveness analyses could be used. For instance, economic methods can estimate costs in quality-of-life measures, such as Disability-Adjusted Life Years (DALY) or Quality-Adjusted Life Years (QALY). Annex 2 contains Table 6 that provides an example of a simple cost-benefit analysis with six steps to estimate the budget required for setting up and maintaining an MCM stockpile. If available, more sophisticated economic analyses can offer reliable estimates, especially for high-volume or high-cost assets. A well-thought-out cost analysis helps weigh the advantages and disadvantages in making the decision to commit the necessary financing to MCM stockpiling. An example of methods used to calculate expected costs is presented in Annex 2- Table 7.

The goals and objectives of the MCM stockpile will help determine the range and quantity of MCM products needed, including any necessary adjustments for different scenarios. They will also guide the types of interventions or actions required to establish and maintain the stockpile, such as training or meetings. The WHO disease commodity package for pandemic influenza offers recommendations on the types of products that are typically considered for inclusion in an MCM stockpile. Inventory management systems are important for setting a minimum threshold for stockpile levels and for tracking and recording products as they move through the supply chain. Reviewing previous expenditures related to routine or MCM stockpiling can provide insights into direct costs, such as those for personnel recruitment and training, procurement, storage (physical or virtual), maintenance, transportation, and distribution. Adjustments may be necessary based on the volume of MCM required to meet the stockpile's goals and objectives. Additionally, indirect costs like waste management, product disposal, expiration schedules, and inventory management software/hardware maintenance can be estimated from historical data.

A sensitivity analysis can help explore the robustness of these cost estimates by varying key assumptions, such as expiration dates, anticipated disease incidence, and stockpile size. This analysis can identify which factors most significantly influence the overall cost. Once an understanding of the overall costs has been developed, it can be useful to compare different stockpiling arrangements—such as physical, virtual, or hybrid models—to determine the most suitable approach. Reviewing various methods for controlling costs, based on known or estimated savings from routine or existing MCM stockpiling, can also provide valuable insights.

Maintaining stockpiles also requires a dedicated and skilled workforce. Countries need to evaluate whether they have the necessary **human resources** in place to manage stockpiles, including inventory management, distribution logistics, and the technical expertise needed to handle complex MCM. Protecting, re-training and re-skilling human resources depends on secure long-term financing to ensure that competencies are sustained to meet routine and pandemic operations, with sufficient personnel to effectively manage the MCM inventory, undertake procurement and analyze data to make the process more efficient. MCM stockpiling personnel must have good problem-solving skills, be able to cooperate with related agencies, and be highly effective in logistics planning for rapid MCM deployment. Therefore, answering a question such as *whether the country has sufficient human resources, or the capacity to train personnel, to manage and maintain MCM stockpiles effectively*, might also involve identifying gaps in current staffing and understanding costs related to planning for recruitment or training initiatives to build the required capabilities.

Adequate **infrastructure** is another critical component. Countries should consider whether they have the physical facilities necessary for the safe storage and management of MCM, including climate-controlled warehouses, secure transportation networks, and robust inventory management systems. The resources required for the management of MCM stockpiles could be integrated or parallel the routine storage and distribution systems of national medical products (e.g. via Central Medical Stores etc.). Some countries may have separate emergency systems for pandemic MCM or have concurrent MCM stockpiles managed by other Ministries (e.g. Defense) or perhaps be linked with state/provincial governments, or hospital networks, or outsource MCM stockpiling to private agencies, or make greater use of virtual stockpiling. Resources for infrastructure might extend to single or multiple storage facilities, and include warehouse management and inventory systems, vehicles, equipment, maintenance, distribution systems, and related quality control testing, or refurbishment of stored products. Therefore, there are several layers to answer a question such as *Is the existing infrastructure sufficient to support the storage and management of MCM, or would significant investments be required?* Understanding infrastructure needs can help determine whether establishing a stockpile is a viable option or if improvements are necessary first.

Finally, the establishment and maintenance of stockpiles require supportive **policies and regulatory frameworks**. Countries need to ensure that there are clear guidelines governing the management of stockpiles, including regulations on procurement, storage, distribution, and the use of MCM during a pandemic. Answering a question such as *Are the necessary policies and regulations in place to support the effective maintenance and deployment of MCM stockpiles?* will include evaluating existing laws and identifying any gaps that need to be addressed to ensure that stockpiling efforts are legally sound.

3. Stockpiles goals and objectives

The risk assessment (encompassing hazard, vulnerability, and capacity analyses) is a foundational step in helping a country decide whether to stockpile pandemic influenza MCM. Beyond this, the assessment should be complemented by considerations such as the potential evolution of a pandemic, the likely demand for MCM at different stages of the outbreak, and the anticipated logistical challenges in distribution and deployment. These factors, along with others like the speed at which a pandemic could spread, the potential for healthcare system overload, and the need for rapid response capabilities, should all inform the development of specific goals and objectives for the stockpiles. Therefore, these periodic and comprehensive risk assessments inform not just a binary question of whether or not to stockpile, but also the level of risk acceptable under different stockpiling methodologies that balance cost and risk.

Once the decision to establish a stockpile is made, goals are essential for guiding various aspects of its setup and ongoing management. For instance, the nature of the goals will determine the quantitative needs the stockpile should meet, thereby influencing the allocation of necessary resources—such as infrastructure, funding, and human resources. If the stockpile primary goal is to ensure rapid response during a pandemic's early stages, it may require a focus on pre-positioning MCM in strategic locations and ensuring quick mobilization capabilities. If, the goal is to ensure strategic national security, the involvement of sectors beyond health, such as the Ministry of Defense, may be necessary to ensure a coordinated, multi-sectoral response. Each objective will shape the strategic approach to stockpile management, including the roles of various stakeholders and the integration of the stockpile within broader national emergency preparedness and response frameworks.

Table 1 presents examples of potential goals that countries may consider when establishing and maintaining MCM stockpiles for pandemic influenza, along with their corresponding objectives and underlying assumptions. These examples are not prescribed in any single global framework but emerge from the common logic of health emergency preparedness and the practical experience of national stockpiling programs. Across contexts, stockpiles are framed as strategic tools to strengthen readiness, enhance resilience, and ensure continuity of response when routine systems are stressed or disrupted. From this perspective, goals such as enabling rapid response, providing surge capacity, mitigating risk, supporting pandemic response, ensuring national security, and promoting cost efficiencies reflect both the operational demands of public health emergencies and the broader principles of risk management.

One frequent goal is of **Supporting a National Rapid Response**, which focuses on the rapid deployment of essential supplies during emergencies. The objective here is to enable quick access and distribution of critical products in the early stages of an outbreak. This goal assumes that normal stock levels within the health system may be insufficient to respond effectively, even to a limited outbreak. To address this, countries might consider strategic placement of stockpiles across different regions to minimize distribution time, along with accurate forecasting and established coordination protocols.

Another goal to consider is **Ensuring Surge Capacity**. The objective under this goal is to expand on existing supplies and infrastructure during periods of extreme demand, ensuring that healthcare facilities can continue to operate effectively. This approach assumes that as an outbreak escalates, the demand for MCM will increase, potentially overwhelming routine supplies. Therefore, achieving this goal would require the ability to rapidly scale up operations, secure additional resources, and ensure that supply chains are resilient and capable of handling sudden increases in demand.

Countries might also consider the goal of **Mitigating Risk**, which aims to address potential supply chain disruptions and ensure the continuity of critical services. The objective here is to maintain operations even if supply chains are compromised. This goal assumes that regular updates to stockpile contents and planning for security risks are necessary and may involve duplicating critical items across multiple locations to avoid single points of failure.

Additionally, the goal of **Supporting the Pandemic Response** could be considered, with the objective of bolstering public health interventions such as vaccination campaigns and disease containment efforts. The assumption here is that stockpiles would need to support a comprehensive pandemic response, adapting to different scenarios and strategies as needed.

The goal of **Ensuring Strategic National Security** might also be relevant for some countries, with the objective of integrating pandemic preparedness into broader national defense strategies. This approach assumes that pandemic influenza poses significant security risks and may require collaboration between military, intelligence, and civilian agencies, as well as secure management of stockpile information.

Countries might explore the goal of **Promoting Cost Efficiencies**, focusing on reducing costs and improving resource utilization through strategic stockpiling. The objective here is to demonstrate economic value, potentially through bulk purchasing agreements, long-term contracts, and partnerships that optimize resources and share costs.

These examples illustrate how different goals, objectives, and assumptions can be tailored to a country's specific context, helping to guide the development and management of MCM stockpiles in a way that aligns with national priorities and resources.

Stockpiling should also be considered in light of broader public health goals. In certain contexts—particularly where global supply is limited—uncoordinated or unilateral stockpiling may unintentionally contribute to inequities in access, securing supplies for some populations while delaying availability for others. Likewise, without effective forecasting, inventory management, and replenishment systems, stockpiling may lead to inefficiencies such as product expiration and wastage. These risks do not inherently undermine the value of stockpiling but underscore the importance of aligning stockpile design and governance with principles of equity, sustainability, and shared responsibility.

Table 1. Goals, objectives and assumptions

Typical goals for an MCM stockpile	Typical MCM stockpile objectives might include	Assumptions
Support a national rapid response	To enable rapid deployment of essential supplies, medicines, and equipment during emergencies.	There will be an immediate need for certain products from initial cases and the assumption is normal stock levels in the health system will be inadequate to respond to even a limited outbreak. The related assumptions are that stockpiled items can be accessed and distributed quickly. This also may imply strategic placement of stockpiles across the country to minimize distribution time. This approach would also need to rely on accurate forecasting of potential emergencies and the types of resources needed. Furthermore, established protocols for coordination between various agencies and organizations for swift deployment, and robust systems for inventory management to keep track of stock levels, locations and expiry dates are critical from an operational point of view.
Ensure surge capacity	To expand on existing supplies and infrastructure during periods of extreme demand or crisis, ensuring that healthcare facilities and responders can operate complex medical equipment (e.g. ventilators)	As an outbreak escalates, there is a need to surge capacities. A stockpile would help meet the surge needs that would imply a higher volume of MCM will be needed as the at-risk population increases. The assumption would be that the quantities routinely available in the health system would not be sufficient to meet these surge needs. Achieving this goal assumes the ability to scale up operations quickly to meet sudden increases in demand. This may also imply having more than the estimated required amounts to account for unexpected shortfalls or higher-than-expected usage. For operational planning it assumes availability of trained personnel to manage and distribute stockpiled resources during a surge. Furthermore, it assumes that supply chains are resilient (agility and diversity of sources) and can be ramped up in times of need. It also implies that there is sufficient funding to maintain higher stock levels and manage surge capacity logistics.
Mitigate risk	To mitigate risks associated with supply chain disruptions, ensuring continuity of critical services and operations.	The outbreak may be severe enough to lead to supply chain disruptions and from the analysis of suppliers and their sourcing there are considerable risks associated with delivery of different products. Assumes regular review and updating of stockpile contents and operations based on evolving risk landscape. Will likely involve a higher degree of planning – including for security risks (theft, damage). It may involve accepting a degree of duplication of critical items across multiple locations to prevent single points of failure.
Support the pandemic response	To support public health interventions, such as vaccination campaigns, disease containment efforts, and distribution of MCM.	These stockpiles would need to sustain the full pandemic response, considering the different MCM strategies that would apply under different scenarios

Typical goals for an MCM stockpile	Typical MCM stockpile objectives might include	Assumptions
Ensure strategic national security	To bolster national security by ensuring readiness to respond to pandemic threats.	This goal would imply a threat assessment that sees pandemic influenza having national security implications and likely will involve collaboration between military, intelligence, and civilian agencies around issues for national defense. These types of stockpiles often imply secure and confidential management of stockpile information to prevent espionage or sabotage.
Promote cost efficiencies	To reduce costs and improve productivity	This goal implies a cost-analysis which would demonstrate cost efficiencies that stockpiling would bring and a regular approach to performing such a cost-benefit or an economic analysis to ensure the stockpiles provide value for money. It may assume cost savings through bulk purchasing agreements and long-term contracts, or procurement coordinated through joint or regional agreements. In addition to various management efficiencies, it may also imply partnerships with the private sector and international organizations for cost-sharing and resource optimization.

Another application of having clear goals and objectives is their role in understanding the potential range of needs, particularly in the quantification of different type of MCM. Some public health goals could be attained by having stockpiling arrangements in place, while others may need to rely on ad-hoc solutions.

Table 2 provides examples of how public health goals and objectives, along with related assumptions, can assist in forecasting and quantifying vaccines and treatment options for pandemic influenza and other respiratory viruses with pandemic potential (adapted from (8)). For instance, to achieve the goal of reducing mortality by treating all severe and critical cases, countries can estimate the percentage of the population likely to be affected. Based on data from the 2009 H1N1 influenza pandemic, approximately 24% of the population was affected (9), with 4% of those cases being severe or critical (9, 10). Therefore, the potential treatment need could be estimated as 4% of the 24% affected population.

As with all such exercises, these calculations require certain assumptions. However, there are limitations, as it is impossible to predict exactly how a novel influenza virus will manifest—meaning that assumptions about its spread and severity are inherently uncertain. The examples provided here use simple methods to aid in these estimations. More detailed modeling methodologies, which incorporate factors such as the effectiveness of MCM and public health and social measures as well as more nuanced epidemiological assumptions, could further refine estimates of the quantities of MCM needed for pandemic influenza.

Table 2. Examples of goals and objectives for pandemic vaccination planning (influenza).

Public health goal		Objectives	Assumptions (refer to WHO source document)
A	Reduce mortality	A-1 Treatment for all severe and critical cases	24% of the population affected ² of which 4% ³ would be severe or critical cases.
		A-2 Treatment for all mild and moderate cases	24% of the population affected, of which 96% ⁴ would experience mild or moderate disease.
		A-3 Vaccine for all people in high-risk groups	Define high risk groups: e.g. obese, aged 65+ years, people with cardiovascular diseases, cancer, diabetes, or chronic respiratory disease
B	Reduce morbidity	B-1 Vaccine to prevent the disease in young and otherwise healthy adults	People aged 20–65 years
C	Maintain critical services	C-1 Vaccinate healthcare workers	Target those in surge capacity
		C-2 Vaccinate people who work in essential services	~2–5% of total population
D	Limit disruption of economic and social functions	D-1 Vaccinate a specified percentage (X%) of the population	X% represents the herd immunity threshold
		D-2 Post-exposure quarantine and prophylaxis (vaccination or therapeutics, as applicable) of close contacts of a case	
E	Protect people in situations of vulnerability/humanitarian settings	E-1 Vaccinate key populations in humanitarian settings	Vaccines can be deployed in certain settings. (e.g. refugees, internally displaced persons)

2 Based on the results of a meta-analysis that estimated the overall age-standardized incidence of 2009 pandemic influenza (9).

3 The mid-range value of severe case rates (2–6%) - analysis of data from the 2009 pandemic influenza (9,10).

4 The mid-range value of mild or asymptomatic case rates (94–98%) - analysis of data from the 2009 pandemic influenza (9,10).

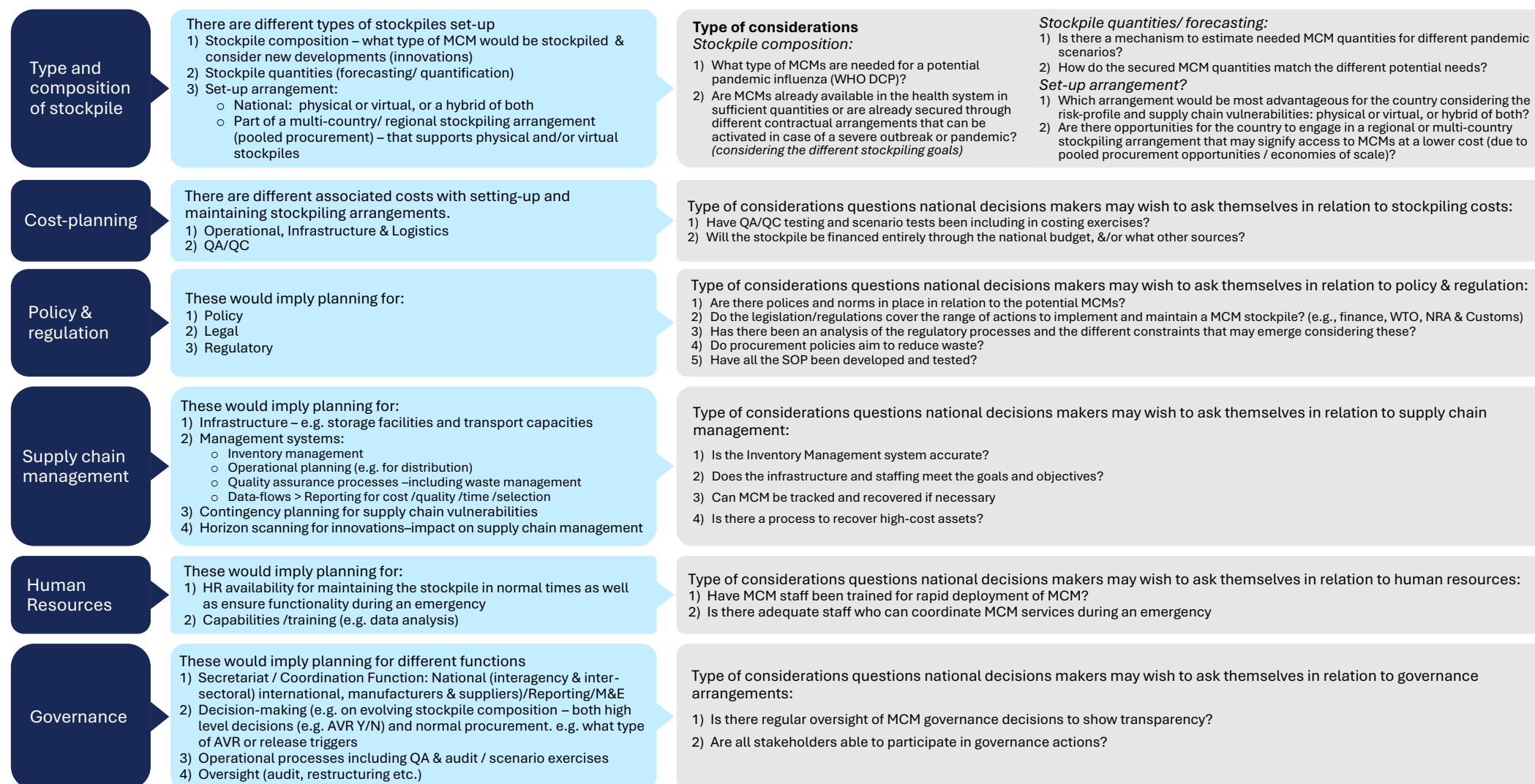
4. Planning considerations and best practices for the set-up and management of stockpiles

Setting up and managing stockpiles for pandemic influenza MCM involves a complex array of considerations that are crucial for ensuring that the stockpile meets the country's public health needs. This chapter outlines the key planning considerations and best practices that countries may want to evaluate when establishing and managing their stockpiles.

The successful setup of stockpiles requires careful attention to several core areas, including the type and composition of the stockpile, cost planning, policy and regulatory frameworks, supply chain management, human resources, and related governance arrangements. Each of these components plays a critical role in ensuring that the stockpile is both effective and sustainable over time.

Figure 3 provides an overview of the key considerations and guiding questions that countries may reflect upon for each of these areas. The following sections delve into these considerations in detail, offering insights and examples that can help inform national decision-making processes.

Figure 3. Planning considerations once a decision has been made to establish stockpile arrangements



4.1 Type and composition of stockpiles

4.1.1 Stockpile composition

When planning the composition of stockpiles for pandemic influenza, it is essential to approach the process with flexibility, tailoring the selection of MCM to the specific context and needs of each country. The WHO's Disease Commodity Package for Pandemic Influenza (5) offers a useful reference, providing a comprehensive list of products that countries might consider including in their stockpiles (see Annex 3). However, the exact composition should be guided by a country's specific risk assessments, capacity, and strategic goals.

The WHO Disease Commodity Package for Pandemic Influenza structures the products according to their use – e.g. to aid surveillance efforts, for infection prevention and control (IPC) activities or for case management.

Surveillance: Effective pandemic response begins with robust surveillance capabilities. Countries may consider including items in their stockpiles that support sample collection and safe transport, such as triple packaging boxes, viral transport medium, and various blood collection tubes (ethylenediaminetetraacetic acid, serum, plain/dry). These tools are vital for the timely detection of the virus and tracking its spread. Additionally, sharps container boxes could be included to ensure safe disposal practices during sample collection, thereby protecting healthcare workers and the broader community. The diagnostic component of a stockpile will need to be adaptable, as the required tests will depend on the specific characteristics of the virus causing the pandemic. WHO guidance, provided during an outbreak, can help countries choose the most appropriate diagnostic tools.

Infection Prevention and Control: Countries might want to consider a broad range of personal protective equipment (PPE) for inclusion in their stockpiles to ensure the safety of healthcare workers and to control the spread of the virus within healthcare facilities. The Disease Commodity Package includes items such as examination gloves, surgical gloves extending to the forearm, face shields, coveralls, N95 or higher-grade respirator masks, surgical masks, scrubs, gowns, head covers, and rubber boots. Other critical items might include protective goggles, aprons, alcohol-based hand rub, bio-hazard bags, body bags, and disinfectants like chlorine. Including such items helps maintain hygiene and safety standards, particularly in high-risk environments.

Case management: For case management, the stockpile could include both aetiological treatments, such as antiviral medications like Oseltamivir and Baloxavir, and supportive care items. The Disease Commodity Package suggests considering items like compound sodium lactate solution, infusion giving sets, paracetamol, oxygen concentrators, flow splitters, nasal oxygen prongs, oxygen tubes, portable ventilators, pulse oximeters, and water for injection. Additionally, the inclusion of standard PPE ensures that healthcare workers are adequately protected while managing cases.

Vaccines: Vaccines present a unique challenge in stockpile planning due to the need for the exact strain of the pandemic influenza virus to be known before production can begin. Countries might explore different procurement strategies, such as advance purchase agreements or regional pooled procurement mechanisms, to secure vaccines. Some countries may also consider maintaining a physical stockpile of vaccines for specific strains, like H5N1, as a precautionary measure. However, given the uncertainty around which strain will cause a pandemic, maintaining flexibility in vaccine stockpiling strategies is advised.

4.2 Forecasting needs

The importance of forecasting as part of a pandemic preparedness strategy is to inform planning for budget allocations to meet public health goals through the timely and equitable delivery of vaccines, antivirals, diagnostics and essential ancillary items such as biohazard puncture-resistant containers for sharps disposal—while minimizing the risk of stock-outs or excess stock that could expire before use.

For routine medical products, forecasting is typically guided by historical consumption data. However, forecasting the needs for pandemic influenza MCM is inherently more complex due to the lack of direct consumption data. In this context, analyzing prior consumption data of similar products, or using proxy data—such as historical data from similar outbreaks or recent pandemics, can provide valuable insights. Comparing this data with distribution records serves as a quality check, helping to validate assumptions and align forecasting with the overall goals and objectives of the stockpile (11).

When conducting forecasting calculations, it is important to consider the entire supply chain, not just the quantities currently on hand. This includes accounting for pending orders, anticipated donations, and any stock that may have been lost, pilfered, or damaged. Additionally, expired products should be systematically removed from inventory to ensure that the stockpile remains viable and effective.

To support these complex calculations, several commercial forecasting tools are available, offering functionalities that can enhance accuracy and efficiency. In general forecasting tools can integrate various data inputs, model different scenarios, and adjust forecasts in real-time as new information becomes available.

4.3 Set-up arrangements

4.3.1 Routine versus emergency stockpiling

The set-up arrangements for routine versus emergency stockpiling of MCM differ in several key aspects, including their purpose, management practices, financing, replenishment strategies, and the types of items stored. Understanding these differences will aid analysis of how best to consider the potential set-up and maintenance synergies.

Purpose and goals: Routine stockpiling focuses on maintaining a supply of medical products that are regularly used to address ongoing health needs, such as vaccines for seasonal influenza or standard diagnostic tests. These stockpiles are typically managed based on historical consumption patterns and are replenished regularly through standard procurement processes. In contrast, emergency MCM stockpiling is specifically designed to address the needs of a potential or ongoing crisis, such as a pandemic influenza. These stockpiles include specialized products and equipment, such as vaccines, antivirals, ventilators, and oxygen supplies, which are essential during emergencies but may not be in high demand under normal circumstances.

Management and coordination: Routine stockpiles are generally managed within established public health frameworks and often fall under the purview of the Ministry of Health or a designated agency. Emergency MCM stockpiling, however, requires a more flexible approach due to the unique and urgent nature of crises. Coordination may still involve the Ministry of Health, but alternative arrangements might also be considered. For instance, a parallel distribution system could be managed by an external agency or a different ministry, such as Defense. In some cases, sub-national authorities (e.g., states or provinces) might establish their own emergency stockpiles. Smaller member states may

leverage existing hospital distribution networks to facilitate the direct supply and staging of pandemic interventions.

Financing and replenishment: Routine stockpiles benefit from regular, predictable funding and replenishment cycles, which are planned and budgeted for as part of routine public health operations. Emergency stockpiles, however, often require distinct financial arrangements. Funding for these stockpiles may come from special emergency budgets or contingency funds. Replenishment strategies for emergency stockpiles are based on periodic risk assessments and updated threat scenarios, and may involve more complex procurement processes to ensure the availability of specialized items when needed.

Types of items stored: The composition of routine stockpiles typically includes products that are used in day-to-day health care and preventive services. In contrast, emergency MCM stockpiles are specifically designed to cover a broader and often more specialized range of items.

Quality and functionality reviews: Both types of stockpiles require regular reviews to ensure the quality and functionality of their contents. Routine stockpiles are typically reviewed based on established schedules aligned with standard health practices. Emergency stockpiles require more frequent and dynamic assessments to account for the evolving nature of threats. This includes updating stockpile contents based on emerging risks, technological advancements, and changes in public health priorities.

4.3.2 Virtual versus physical stockpiles

MCM stockpiling strategies encompass virtual and physical stockpiles, with many countries opting for a hybrid model that combines elements of both (12). The choice between these approaches is influenced by factors such as cost, logistics, and the ease of access to target populations. An effective stockpiling strategy typically integrates both physical and virtual elements to ensure supply chain resilience and readiness, tailored to the specific needs and goals of the country. Table 3 provides an overview of the advantages and disadvantages of different stockpiling approaches.

Virtual stockpiles: Virtual stockpiles are those where MCM are not physically stored by the country but are instead held by manufacturers or other partners. This approach can lead to significant cost savings by eliminating the need for physical storage facilities and associated maintenance costs. Virtual warehousing allows for a single, unified view of inventory across multiple locations, improving efficiency and reducing the risk of stockouts. It offers flexibility, as the stockpile can be easily scaled up or down based on demand. However, virtual stockpiles also come with certain risks and disadvantages. The country becomes dependent on manufacturers to produce and deliver essential MCM during a crisis, which can lead to delays if supply chains are disrupted. A best practice is to build delivery obligations into the contract, with specific clauses on timelines, penalties and other such details. Organizing the supply chain and coordinating with multiple stakeholders during an emergency can be complex and time-consuming, potentially slowing down the decision-making and deployment process. Additionally, virtual stockpiles often maintain minimal inventory, reducing the buffer available to respond to unexpected surges in demand.

Physical stockpiles: Physical stockpiles involve the storage of MCM in designated facilities, often managed by a government agency, that are ready for immediate deployment during an emergency. These stockpiles can range from multi-country or regional facilities to national or sub-national warehouses. Physical stockpiling provides several key advantages, including the immediate availability of products and resilience against external supply chain disruptions, ensuring that critical MCM are available when needed. Moreover, they allow for faster decision-to-deployment times, which can be crucial in the early stages of a pandemic. However, physical stockpiles come with high costs,

associated with creating and maintaining storage facilities, ensuring quality control, providing security, and conducting regular inventory management. Over time, these costs can be significant, especially if the stockpile needs to be replenished or updated regularly to account for changes in risk assessments or emerging threats.

Hybrid approaches: Many countries find that a hybrid approach, combining both physical and virtual stockpiles, offers several advantages. This strategy allows for the immediate availability and security of physical stockpiles, complemented by the flexibility and potential cost efficiencies of virtual stockpiles. For instance, a country might maintain a core inventory of essential MCM in physical storage while relying on virtual arrangements for less critical items or for rapid scale up during a crisis.

When considering whether to adopt virtual, physical or hybrid stockpiling, countries should make their decisions according to their stockpiling goals and objectives that considers their needs, risks, and available resources – including enabling infrastructure—such as reliable electricity for cold chain, transport access to remote areas, and regulatory preparedness among others. Engaging in discussions with stakeholders and particularly industry partners can identify the potential opportunities and limitations, to provide a context to inform the decision-making process and guide the selection of the most suitable approach. In lower-income countries, this may include exploring innovative, low-cost hybrid models or regional agreements to overcome infrastructure, financing, and procurement limitations.

Table 3. Physical versus virtual stockpiles

Physical stockpile	Virtual stockpile
Advantages • Safe storage • Temperature Control • Product Integrity and Safety • Inventory Management • Always available for dispatch • Faster decision to deployment times • Regional Resilience	Advantages • Cost efficiencies • Security • Flexibility; Easy to scale up or down based on need • Ease of use; Managing virtual stockpiles is straightforward • Finances are allocated when a crisis occurs • Maintenance free
Disadvantages • Cost of creation • Quality control • Maintenance and security	Disadvantages • Dependence on companies to manufacture essentials at the time of crisis • Organizing the supply chain at the time of crisis • Slower decision to deployment times • Minimal inventory, reduced buffer

Adapted from (11).

4.3.3 Procurement

Following on from the decision on the format of the stockpile – whether virtual, physical or hybrid, countries should consider also the type of connected procurement approaches. Several key considerations must be addressed during the procurement process to ensure that the necessary medical products are available in a timely manner, are of high quality, and are obtained at an affordable cost.

Mapping expertise and relationships with vendors

Understanding the market landscape and establishing clear relationships with vendors can help in managing procurement risks. Countries might find it useful to map the expertise of their vendors and document interdependencies between vendors, which could be especially important when dealing with virtual or hybrid stockpiles where multiple suppliers might be involved across different locations. This approach may help in identifying the most reliable and qualified suppliers, whether for maintaining physical inventories or for securing virtual commitments from manufacturers.

Ensuring high-quality products

The procurement of MCM involves more than just acquiring products; it also involves ensuring that these items meet rigorous quality standards, particularly for physical stockpiles where maintaining product integrity over time is crucial. Specifications for MCM may include compliance with Good Manufacturing Practices (GMP), product registration with a high-level National Regulatory Authority (NRA), WHO prequalification, and certifications such as ISO compliance. For virtual stockpiles, where the focus might be on securing agreements for rapid production and delivery, these quality standards remain important, with an emphasis on ensuring that suppliers can meet these standards quickly during an emergency.

Adopting flexible and transparent procurement processes

In recent decades public procurement has moved toward simpler, flexible processes that increases competition, transparency, and accountability (e.g. minimizing low-value repeat purchases, increased use of economies of scale, and reduced transaction and transport costs). Many countries have instituted national online systems for pre-registered vendors where their legal status, credentials, and experience must be disclosed to be eligible to apply for public tenders. e-GP (e-government procurement) ranges from publishing of bids through a centralized web portal, to e-tendering, or as total end to end e-procurement including contract and payment management (13,14).

Engaging Procurement Agents

Government agencies or other organizations are able to outsource procurement of goods and services to procurement agents (PA) under contract. The PA role involves working closely with suppliers to meet their contractual obligations to supply goods or services in terms of quality, cost, and timing. With respect to MCM, countries may wish to consider including as a condition of the PA contract to map vendors according to product type, lead times, and other key characteristics. This is important for either physical or virtual stockpiles, where the timely availability of products depends on the manufacturer or supplier to meet extreme demands in very short timeframes. PA use a variety of methods to source products and may issue open or select tenders, maintain their own discrete supplies, or use preferred providers whose credentials will depend on the specifications set by the buyer. These specifications could mean for MCM (where relevant): GMP compliance, product registration with a ML3-ML4 rated NRA, WHO PQ, sterile manufacture, quality control (QC) certificate, batch testing, lot release, ISO compliance, industry code, environmental or other specified standards etc.

Conducting market analysis

Given the ongoing advancements in MCM research and development, it can be beneficial for countries to stay informed about new and innovative products that could potentially be added to stockpiles, particularly when managing virtual or hybrid models. Regular market analysis offers a way to stay updated on developments in the pharmaceutical and medical device sectors, helping to identify which products might be best suited for physical stockpiles and which could be more effectively sourced through virtual agreements. For instance, a country might choose to physically stockpile critical

items like ventilators and antivirals, while relying on virtual arrangements for vaccines that need to be produced based on the specific strain of the virus. National decision-makers might consider diversifying the types of products stockpiled as innovative products become available.

The MCM market is influenced by various factors, including advancements in technology, shifts in workforce dynamics, and lessons learned from previous pandemics like COVID-19. AI and analytical software can help a more rapid understanding of value chains, greater movement toward autonomous supply chains, reducing wastage and improving production costs (return on investment). It might be worthwhile for countries to consider how these trends impact their procurement strategies (15,16), particularly when choosing between physical, virtual, or hybrid stockpiles. For instance, advancements in AI and supply chain analytics could improve the efficiency of managing virtual stockpiles by predicting demand and optimizing delivery schedules.

4.4 Cost related planning

Building on the cost considerations discussed in Chapter 2.3, that looked at estimating the costs associated with setting up and maintaining stockpiles, this section addresses some key aspects of stockpile maintenance and cost management.

Stockpile maintenance

Maintaining the quality and functionality of MCM stockpiles involves routine testing and inspection of stored items. Regular QC checks help confirm that supplies meet applicable standards and remain effective. To support this, it is important to allocate regular budgets for QC testing that also includes testing of the functionality of equipment, and compliance with Good Storage Practices to prevent issues like deterioration.

Containing Costs

MCM in these stockpiles may be used for both frequent, lower-impact events (such as seasonal influenza outbreaks) and less frequent, high-impact situations (like a pandemic). Managing costs effectively can involve strategies such as rotating stock to integrate expiring items into the routine supply chain while replenishing the stockpile with new items (17).

For high-volume, low-risk items (like PPE), bulk purchasing can often be cost-effective approach and help avoid the higher costs of emergency purchases. Additionally, pooling resources through multi-country procurement can enhance access to MCM and support cost savings. Strategic partnerships and forecasting and procurement tools may help minimize supply chain disruptions and streamline procurement, avoid stockouts and contribute to cost efficiencies (18).

Having pre-negotiated contracts with suppliers can spread the risk of maintaining a steady supply of goods, particularly during the early stages of a pandemic. Efficient inventory management is key, including policies for rotating stock and handling unused or excess MCM through sale, transfer, or donation. Effective waste management helps control costs by freeing up space in storage facilities. Procurement plans should consider durability and environmental factors, as longer-lasting, eco-friendly products can reduce costs over time, while ensuring transparent procurement practices helps prevent waste, ensures that suppliers meet quality standards, and safeguards public health by reducing the risk of unusable products.

4.5 Policy and regulations

4.5.1 Policy articulation

The formulation of policies for MCM stockpiling involves creating a structured approach to govern the acquisition, storage, management, and deployment of medical supplies intended for use during health emergencies. These policies are essential for ensuring that MCM stockpiles are effectively maintained and utilized to protect public health during crises.

Government agencies work to craft policies that outline the principles and procedures for MCM stockpiling. This includes defining roles and responsibilities, setting objectives, and creating guidelines that reflect national priorities and public health goals. These often involve consultation with various stakeholders, including health experts, emergency planners, and legal advisors, to ensure that the policies are comprehensive and aligned with broader public health strategies.

Effective policy development requires engaging with national and sub-national authorities, as well as potential international partners. This collaborative approach helps to build consensus and ensures that the policies are feasible and address the needs of all involved parties. Engagement may include discussions on pooled procurement, sharing of MCM, and coordination with international health organizations.

Policies must be formally approved by relevant governmental bodies. This often involves a review process where the policies are assessed for alignment with existing laws and regulations. After approval, clear communication is essential to ensure that all stakeholders, including health agencies, emergency responders, and the public, understand the policies and their roles in implementation.

Policies for MCM stockpiling should be regularly reviewed and updated to reflect changes in public health priorities, advancements in medical research, and evolving emergency response needs.

4.5.2 Legal considerations

Legal obligations related to MCM stockpiling generally align with those for routine stockpiling and include compliance with national laws and regulations concerning customs, market authorization, and exemptions or waivers where applicable. It is also important to adhere to environmental regulations regarding disposal and waste management.

The legal framework governing the infrastructure, storage, maintenance, and procurement of MCM stockpiles might be covered by existing legislation or may be linked to separate emergency laws (19). For instance, during events like the COVID-19 pandemic, innovative or unregistered MCM sourced for stockpiling may require special attention to legal provisions, including liability, indemnity, or compensation agreements with manufacturers to ensure public protection.

Conformance is also necessary with Intellectual Property rights, waivers, Trade-related aspects of Intellectual Property Rights and other World Trade Organization agreements, or through Mutual Recognition Agreements with trading partners.

To facilitate the smooth importation or export of emergency supplies means understanding the types of documents reviewed by Customs, and being aware of any special conditions or emergency provisions that might apply to MCM. These include import or export certificates with stated exemptions (if any); transport documentation (bill of lading; tax; packing list; insurance; etc.) receipts for taxes or other fees, and documentation issued by the relevant NRA for medical products or medical equipment, such as market authorization; GMP certificates; and/or test certificates, (e.g. lot release or batch certificates).

Staff who are responsible for procurement of MCM may be interested in completing IMPortation And Customs Clearance Together - IMPACCT online courses to improve their skills, delivered for example by the United Nations Children's Fund, Médecins sans Frontières and others, designed for individuals in United Nations agencies, nongovernment and government agencies, and the private sector who are involved in the importation of humanitarian goods (20).

4.5.3 Regulatory considerations

Administration and enforcement of laws and regulations for the safety, quality, and efficacy of medical products is predominantly the responsibility of the NRA which usually is an agency within the Ministry of Health. Effective MCM stockpiling requires adaptable regulatory frameworks that ensure safety and quality, even in emergency situations. This flexibility allows for the rapid introduction of both existing products with marketing authorization and new products, such as pandemic influenza vaccines, which may be subject to advance procurement but not yet have formal authorization.

Countries should leverage the WHO global benchmarking tool to assess their current regulatory capabilities. The WHO global benchmarking tool provides ratings from ML1 (minimum) to ML4 (highest) and helps identify any gaps in the regulatory system, ensuring that countries are prepared to regulate MCM effectively. In the interpandemic period, efforts should focus on establishing clear pathways for the expedited authorization of new MCM during health crises. These pathways may include independent review of the full dossier, emergency authorization, and reliance or recognition procedures (21).

After authorization for use is granted, the NRA is also usually responsible for reviewing various documentation to ensure ongoing compliance and upholding of quality standards. This include confirming that MCM products are properly labeled or relabeled in the event of extension of shelf-life as well as reviews to verify the quality control and assurance for respective lots of products.

In emergency situations, there may be provisions for waiving certain fees or taxes related to customs and NRA procedures. Ongoing collaboration with the NRA is important for managing issues such as product recalls, withdrawals, and disposals, ensuring that all actions align with regulatory requirements.

Regardless of the source of the MCM, NRAs, along with national specialized programs (e.g., immunization programs) and other relevant stakeholders, should develop robust strategies for monitoring MCM safety. These strategies should address risk communication and management, including conducting post-authorization studies to ensure safety across various populations.

4.6 Supply chain management

4.6.1 Understanding supply chain dynamics

Effective management of MCM stockpiles relies on the seamless integration of both internal and external supply chains. The internal supply chain, usually managed within the health system is generally more governable, whereas there is less control over the external supply chain of suppliers and manufacturers, including being able to control for potential disruption risks.

Supply chain resilience is a key concept that combines diverse sourcing with agility to adapt to new circumstances. This approach helps supply chains withstand the shocks and demands experienced during crises. For internal supply chains, resilience involves:

- **Cost-effective and high-quality MCM:** ensuring that MCM are not only cost-effective but also meet high safety and quality standards.
- **Agility:** having efficient managerial cooperation and accurate data to make rapid decisions during a crisis such as the efficient movement and re-routing of MCM when necessary.
- **Diverse sourcing:** minimizing risks by planning for a range of suppliers and establishing backup arrangements through direct agreements or regional procurement hubs.
- **Healthcare staff readiness:** ensuring sufficient trained healthcare staff are available to operate complex MCM assets if they are released from the stockpile (e.g. ventilators, during a surge) (22).

Improving resilience in the internal supply chain may involve focusing on responsiveness, managerial coordination, and building strong relationships with emergency suppliers or regional procurement hubs (23, 24).

During an emergency such as a pandemic, manufacturers will likely face disruptions in accessing ingredients, components, or raw materials. The pharmaceuticals, healthcare, and e-commerce sectors faced exceptionally high burdens during the COVID-19 pandemic and had to rapidly reconfigure supply chains to meet an intense surge. Successfully responding to extreme demands was possible in part because of a company's entrepreneurial orientation and adoption of a digital supply chains that helped navigate disruptions and respond faster to volatile supply and demand. The COVID-19 pandemic accelerated pre-existing issues in the global supply chain and brought priorities such as visibility, resilience and digitization into greater focus for the manufacturing sector (25). Recent studies examining the COVID-19 response (24-27) looked the features of companies that point to dynamic capabilities making them more resilient at the time of a pandemic. These 'hyper-agile' organizations demonstrated: (i) data analytical capability - the extent a supplier applies data analysis to drive decision making to facilitate visibility, information sharing, and transparency, (ii) supply chain integration - the level to which a company makes use of their core competencies and collaborates with their own suppliers and networks to re-organize and improve the supply chain, (iii) entrepreneurial and making use of market intelligence to improve production (26-28).

Therefore, in choosing supplies, a country may want to ascertain how adept suppliers will be to be able to prioritize short-term goals, address supply chain uncertainties, and effectively respond to sudden and extreme demands. Mapping suppliers may require analyzing organizational characteristics such as industry type, experience with disruptions (including their causes), organizational size (number of employees), public or private status, years in operation, and geographical location(s) (29, 30) (see also Annex 4).

4.6.2 Storage facilities and infrastructure

Storage facilities and related infrastructure may coincide with routine, regional or companion stockpiles or set-up as independent MCM sites. Being able to make use of existing inventory management technology will aid a coordinated approach for rotation or re-routing of MCM, as well as for retrieval of products for inspections, scheduled quality control testing, or to locate sub-standard products and those for recall. As for routine stockpiles, Good Storage Practices / Good Distribution Practices guidelines should be followed for storage, handling and distribution that includes having secure IT systems to maintain confidentiality of data related to stockpile locations and their contents (31).

Distribution and logistics

Distribution plans should also conform with the goals and objectives developed for the MCM stockpiles, supported by regular training and simulation exercises for all staff to ensure they are prepared for real-world events. Having clear lines of communication and coordination is essential

among stakeholders within the emergency response network (related government agencies, animal health, humanitarian agencies, private sector partners etc.) for delivery of supplies and equipment, especially for people in hard-to-reach locations. Engagements with customs and air/land/sea transportation authorities, the national disaster coordinating committee (which may have access to national assets to support MCM emergency deployment), and private sector actors may help address gaps or provide surge capacity for storage and transport.

The goals and objectives should emphasize equity in the allocation of MCM and planning should account for how scarce resources should be allocated, especially if shared stockpiles may be servicing multiple jurisdictions, and this process should be reflected in distribution plans.

The MCM risk assessment should be regularly tested to ensure local systems including access to a (centralized) stockpile can be fully exploited before drawing on regional stockpiles if borders become closed. Prior liaison and cooperation with the NRA and Customs for applicable waivers, or exemptions will also help expedite movement of supplies across borders during an emergency. Planning should include mechanisms to manage near-expiry supplies, reduce wastage, and address audit and liability concerns. Post-crisis, the recovery of high-value equipment or medical products should also be part of the planning process that describes the coordinated recovery and return to the inventory, for example ventilators after decontamination and a maintenance checks (32).

4.6.3 Tracking MCM

The limited ability to track products caused major challenges during the COVID-19 pandemic particularly during the high initial demand for PPE rapidly followed by critical PPE shortages for frontline workers (30). Traceability of all MCM products is essential to enable reliable reporting of stock levels and expiration dates to prevent wastage, and if requested by the NRA, to help monitor and report adverse events, and provide support for their recall or withdrawal actions.

In practical terms effective communication plays a vital role to inform staff in different locations of the movement or re-routing of MCM assets so that facilities know where supplies are enroute, and especially when the MCM will arrive. Locating lost, misplaced or pilfered equipment perhaps using radio frequency identification technology for example provides a potential solution that also helps with accurate inventory, replenishment, and invoicing (34, 35).

4.7 Human resources planning

Best practice for MCM stockpiling demands a multidisciplinary skill set, with a solid understanding of the procurement, handling and distribution of MCM, and demonstrated capacity to work under pressure, address challenges, minimize disruption risks, and facilitate timely deployment. The readiness of the stockpile falls on the ability of staff to cooperate and communicate not only between sites but also with national and international partners, including for example, by working on joint procurement initiatives, using data analysis to provide an insight to resolve issues related to procurement, logistics and inventory management, and to reinforce emergency procedures through regular (annual) drills.

4.8 Governance

4.8.1 Governance structures

The threats prompting the establishment of an MCM stockpile affect both national security and public health, and require complex coordination likely involving multiple jurisdictions, several levels of government and emergency response agencies to ensure the effective management and deployment of MCM during a pandemic.

Some countries may choose to outsource the management of MCM stockpiles to wholesalers or to civil society partners. However, a central coordinating body, often a national health authority or a dedicated emergency preparedness agency, usually oversees these stockpiles. This body ensures that all activities related to the MCM stockpiling are coordinated and aligned with national health strategies.

Firstly, a policy and legal framework is essential to define the roles and responsibilities of stakeholders (such as government agencies, health authorities, and private sector partners) that is in step with national emergency planning, response, and recovery. Often a sub-committee is convened to review the goals and objectives set for the stockpiles, and examine in more detail the operations (procurement, storage, maintenance, distribution and deployment) as well as related expenditures and resources. This group is also typically responsible for advocating for ongoing financial support from the government for the stockpile.

The composition of MCM committees vary but might include a range of stakeholders directly and indirectly involved with MCM (government agencies, suppliers, manufacturers, community representation, communicable disease experts) to support both the governance and regulatory processes to ensure equitable access to cost-effective, safe, and appropriate MCM during a pandemic emergency.

4.8.2 Monitoring and Evaluation

Monitoring systems to assess pandemic preparedness, readiness and functioning of MCM stockpiles may involve electronic systems as well as protocols and procedures that clearly state the expected performance. Post-emergency activities should foresee in-depth assessments to discover any additional actions that are needed to improve the response, which would inform the updating of existing Standard Operating Procedures and related policies/protocols.

4.8.3 Key performance indicators

Key performance indicators (KPI) help to provide insights into stockpile performance and to monitor and mitigate risk. Depending on the phase of the pandemic, the component areas to measure might cover: (1) accuracy of inventory management systems; (2) timely access to MCM; (3) quality and efficacy of stockpiled items; (4) optimized costs.

The KPI are subject to change depending on stockpile objectives, regulatory requirements, or emerging public health threats. Regular QC and/or functionality testing of MCM in the stockpile should be established for medical products/equipment according to applicable standards and/or manufacturers specifications to ensure all items in the MCM stockpiles remain ready for deployment.

To support the usability of KPI, these should correlate with targeted area for measurement, with a description of the measurement, a definition of the KPI that is expressed according to a predefined target over a specified timeframe (e.g. annually or quarterly). Examples of KPI are provided in Table 8.

Table 4. Key performance indicators examples

Components to measure	Key Performance Indicators (KPI) examples
(1) Medical Countermeasures (MCM) Inventory Management systems: Accurate record of stock visibility, quantities, shelf life and expiry - Inventory Turnover Rate as the frequency at which stockpiled items are used and replenished - Stockout is defined as falling below pre-defined minimum volume - Stockout may indicate: - Inventory count discrepancies - Incorrect forecasting / inaccurate reporting - Slow replenishment - Variations in lead-time	(e.g. meets pre-defined % per quarter) - Accuracy (%) of the stock records compared to the physical inventory - Quantity and types of MCM in stock by item/category (%) - Shelf life and expiry by item/category (% retained) - Expired MCM by item/category that require disposal (% expired). - Inventory Turnover Rate by item/category (%) - Frequency of stockouts (per item /category)
(2) Timely deliveries Distribution and logistics: - Compliance with storage conditions - Compliance with distribution practices - MCM are delivered to specified locations within set timeframe for emergencies - Suppliers deliver MCM according to terms of contract	- Storage locations meet Good Storage Practices - Distribution meets Good Distribution Practices - Average time to distribute MCM to the point of use during an emergency /or simulation exercises compared to pre-defined target in hours (hours) - Reliability of suppliers in delivering MCM on time and in the correct quantities stated in contracts (%)
(3) Cost The average direct cost incurred for procuring, storing, and distributing each category of MCM	The average annual direct cost incurred for each item/category of MCM - Procurement (USD/year) - Storage (USD/year) - Distribution (USD/year)
(4) Safety and Quality According to a predefined QC plan, the percentage of MCM that meet regulatory and quality standards through regular monitoring and testing of QC and/or functionality.	According to medical product/equipment specifications MCM meeting predefined safety /quality standards by item/category (%)

Periodic reviews and regular simulation exercises

Periodic reviews ranging from Intra-Action Reviews, After Action Reviews, independent reviews, and Joint Evaluation Exercises, together with simulation exercises (tabletop or actual) help ensure procedures are operational for an effective response (36-38).

All staff involved in MCM planning should be well-versed in broader emergency protocols, understand their specific responsibilities, and be trained to operationalize these activities through tabletop and simulation exercises. Whenever possible, cross-border drills should be conducted to provide valuable practical experience. Ultimately, distribution from the MCM stockpile must continue without interruption throughout a pandemic, in coordination with and supported by other essential emergency services.

5. Conclusions

This document provides a comprehensive framework for establishing and managing stockpiles of MCM for pandemic influenza. This guidance therefore could support countries in preparing for and responding to pandemic threats by ensuring that stockpiles are effectively designed, implemented, and maintained. The key considerations highlight the necessity of understanding hazard likelihood, vulnerabilities, and national capacities that would warrant establishing resilient stockpiles capable of withstanding and adapting to the challenges posed by pandemics.

Effective stockpiling management requires an integrated approach that encompasses the goals and objectives of stockpile creation, which should align with broader pandemic preparedness and response plans. By addressing these considerations, countries can enhance their readiness and capacity to deploy MCM efficiently in emergencies.

Planning and best practices play a key role in the successful setup and management of stockpiles. The document outlines essential elements such as stockpile composition, procurement strategies, cost planning, and adherence to policy and regulatory frameworks. Emphasizing these aspects ensures that stockpiles are not only well-suited to meet the demands of a pandemic but also compliant with legal and regulatory requirements. Furthermore, effective supply chain management, including understanding supply chain dynamics and infrastructure requirements, is crucial for maintaining the continuity and reliability of stockpiles.

Governance, including approaches to monitoring and evaluation are central to ensuring that stockpile management practices are clear and efficient and continuously refined and improved. Best practices stress the importance of good coordination with related emergency response agencies including animal health and NRA/Customs, as well as maintaining strategic relationships with civil society and other local stakeholders that will add to the overall effectiveness of stockpile management.

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Annex 1. Emergency preparedness and response national stockpiling capabilities

Table 5. WHO Health emergency preparedness, prevention, response, and resilience (HEPR) Five Cs - Relevant sub-capabilities at the national level for MCM stockpiling

Core health emergency components (HEPR 'Cs')	Recommended HEPR sub-capability
(3) Safe and scalable care	3.1 Scalable clinical care during emergencies
	3.1.3 Stockpiles and supply chain for clinical care during emergencies
	• Definition of a list of essential medicines, supplies and prioritized investigational products for priority hazards • Forecasting and mapping of needs for essential medicines and supplies through established inventory management and control systems at facility level based on capacity assessments • Optimization of supply chain and responsive stockpile management processes to ensure equitable access to essential medicines and supplies, including processes for replenishing supplies post-response • Ensuring safe and equitable deployment of investigational products through development of protocols and appropriate workforce training
	3.2 Protection of health workers and patients
(4) Access to countermeasures	3.2.2 IPC in the context of Health Emergencies
	• Development of responsive stockpiling processes to ensure adequate supply of PPE and IPC equipment
	4.3 End-to-end health emergency supply chains
	4.3.5 Resilient logistics and distribution
	• Strategic stockpiles of medical countermeasures are established and provide for rapid response, prevention, response, and stopgap needs.

Annex 2. Cost analysis resources

Table 6. Six steps for a simple cost benefit analysis

Simple cost-benefit analysis – Example steps	
Step	Examples
1 Define MCM stockpile goals and objectives goals and objectives will evolve during different pandemic phases	based on known and/or historical epidemiological data and estimated impact
2a Identify interventions	- training of MCM personnel in inventory management software - amendments to national laws and regulations - regular coordination meetings with regional partners in emergency preparedness
2b Identify appropriate MCM	- based on WHO disease commodity package (pandemic influenza) - based on historical procurement - based on surveillance threat, at-risk populations
3 Estimate costs of MCM / interventions based on routine or MCM stockpiling expenditure	- from known or historical budget allocations/expenditures where possible - estimate of volumes of MCM needed to satisfy goals/objectives - using proxy MCM consumption data from neighboring countries combined with existing MCM distribution data (if available)
4 Sensitivity analysis - test assumptions by changing parameters of MCM / interventions	MCM - adjust the predicted morbidity /mortality estimates - adjust expiry dates - adjust volumes procured for storage - adjust according to expected lead times Intervention - adjust according to time to complete procurement (with and without intervention) - adjust according to delivery schedule - adjust according to full time /part time personnel
5 Compare cost of stockpiling alternatives	- virtual stockpile - outsourced managed stockpile - procurement agent - regional partnership - multi-country partnership
6 Control costs	- Assess ways to control costs to minimize waste, eco-friendly / sustainable products - stock rotation; sale; auction; transfer; or donation - waste management: reduced packaging - e-procurement - bulk purchase - pre-negotiated contracts - refurbished equipment

Table 7. Types of cost analysis

Example of different cost analyses of air purifiers vs. Heating, Ventilation, and Air Conditioning HVAC (*Source (39)*)

Types of Comparative Evaluation for Covid-19 Policies and Interventions*			
Analysis Type	Outcomes Reported	Challenges and Level of Aggregation	Example - air purifier vs. HVAC system
Cost-benefit	Costs and monetized benefits	Optimal choice has highest monetary value; reasonable estimates are required to monetize all outcomes	Assessing the net monetized benefits of adding an air purifier vs. updating an HVAC system
Direct cost	Direct costs, potentially indirect costs	No measures of effectiveness are included; approach is best suited for evaluating interventions with similar effectiveness	Comparing costs between adding an air purifier and updating an HVAC system to improve ventilation
Comparative-effectiveness	Common unit of effectiveness (e.g., cases averted, deaths averted)	No measures of costs are included; outcomes typically aren't aggregated into a single measure	Comparing expected Covid-19 cases averted for adding an air purifier vs. updating an HVAC system
Cost-consequence	Costs and various measures of effectiveness (e.g., deaths averted, hospitalizations averted, direct and indirect intervention costs)	Outcomes are reported separately with no aggregation; a dashboard showing outcomes for each option is provided for decision makers	Comparing costs and expected COVID-19 cases averted for adding an air purifier vs. updating an HVAC system
Cost-effectiveness	Costs and a common unit of effectiveness, often a metric that combines length and quality of life into a single measure (e.g., QALY or DALY)	Relative advantages are quantified for each option; optimal choice depends on willingness to pay for health	Assessing the cost per QALY saved for adding an air purifier vs. updating an HVAC system
Extended cost-effectiveness	Costs, a common unit of effectiveness, and an outcome related to health equity (e.g., new cases of poverty averted)	Equity-related outcomes are reported separately with no aggregation; a dashboard showing outcomes for each option is provided for decision makers	Assessing the cost per QALY saved, and per other outcome (e.g., new cases of poverty averted), of adding an air purifier vs. updating an HVAC system
Distributional cost-effectiveness	Costs and equity-weighted health outcomes (weights based on societal aversion to inequities in the distribution of health)	Optimal choice is based on equity-weighted outcomes and costs; requires reasonable estimates to monetize health outcomes and quantitative estimates of societal preferences regarding the equitable distribution of health	Assessing the equally distributed equivalent (a metric that aggregates the distributions of health and cost outcomes) of adding an air purifier vs. updating an HVAC system

*QALY quality-adjusted life-year, and DALY disability-adjusted life-year. HVAC - Heating, Ventilation, and Air Conditioning

Annex 3. WHO Disease Package for Pandemic Influenza

Type of overall activity	Type of product categories	Products
Surveillance	Sample collection	• Triple packaging boxes • Viral Transport Medium • Tubes, blood collection, ethylenediaminetetraacetic acid • Tubes, blood collection, serum • Tubes, blood collection, plain/dry • Sharps container boxes
	Diagnostics	The WHO will advise on the selection of tests on a case-by-case basis as determined by a specific event.
Infection Prevention and Control	IPC - PPE	• Gloves, examination • Gloves, surgical, length to forearm large (longer than examination gloves) • Face shield • Fit Test Kit • Coverall • Face mask, particulate respirator, grade N95 or higher • Mask, surgical • Scrubs, tops • Scrubs, pants • Gown • Head cover • Boot, rubber • Goggles protective • Apron • Alcohol-based hand rub • Bio-hazard bag • Body bag • Chlorine
	Health Care Facilities	
Case Management	Aetiological Treatment	• Oseltamivir • Baloxavir
	Supportive Treatment	• Compound Sodium Lactate Solution • Infusion giving set • Paracetamol • Oxygen concentrators • (Oxygen concentrator) Flow splitter • Oxygen prongs, nasal, non-sterile, single use • Oxygen tube, extension • Portable ventilator • Pulse Oximeter • Water for injection
	Standard PPE	• Gloves, examination • Gloves, surgical, length to forearm large (longer than examination gloves) • Face mask, particulate respirator, grade N95 or higher • Fit Test Kit • Gown

source: WHO Disease Commodity Packages - Pandemic Influenza (5)

Annex 4. Examples of how to structure suppliers mapping

Table 8. Mapping sources of MCM (example)

MCM Product category	MCM Product type	Date of Inventory	MCM Stockpiled volume (units)	Location(s)	Site 1 central	Site 2	Site 3	Site 4	N/A
PPE	Face Mask N95	1-Jun-24	1,000,000	All sites	700,000	100,000	100,000	100,000	
Source Nat /Int (Sup) or (Man)	Countries of origin	Company name	Replenishment schedule and volume (units)	Contract type	Lead time	Urgent estimated need (units)	Lead time	Completed report on urgent capabilities	Completed report on Quality Assurance
International manufacturer	Country 1	ABC P/L	N/A	NIL	6 weeks			Yes Audited June 2024	Yes Audited June 2024
National Manufacturer	N/A	N/A							
National Supplier	Country 1 Country 2	DEF P/L XYZ town	Annual 200,000 units	Recurrent to 2030	4 weeks	500,000	2 weeks	Yes Audited June 2024	Yes Audited June 2024
National Supplier	Country 3 Country 4	HIJ P/L KLI town		Emergency		100,000	1 week	Yes Audited June 2024	Yes Audited June 2024
National Supplier	Country 4	OPP P/L QLE town	Annual 100,000 units	Alternate	4 weeks	100,000	2 weeks	Audit due Dec 2024	Audit due Dec 2024

Annex 5. Methodological approach

The methodological approach consisted of the following steps.

A literature review was completed of publications sourced from WHO and other databases including PubMed, Google Scholar, PLOS one. These sources were examined to help establish best practice for MCM stockpiling methods, that includes defining the type, size, and contents of the MCM stockpiles, as well as suggestions for appropriate stockpile set-up and management according to different scenarios applying to pandemic influenza.

Table 9. Literature search - inclusion and exclusion criteria

Question components	Criteria	
	Inclusion criteria	Exclusion criteria
Type of subject matter	All relevant subjects discussed in connection to pandemic preparedness and response.	Sub-national, hospital, or non-emergency setting
Type of emergency	While the consideration document is targeting pandemic influenza stockpiling, the literature search was not restricted to that, to identify all best practices related to stockpile management.	Non respiratory related Non pandemic
Study designs	Any quantitative or qualitative comparative study, including RCT, cohort, case-control, case series with historical control, time series. Published between 2009-present. Reported in English	Pre 2009

The literature search was based on key word search and/or citations in published works with a similar focus. The results were merged into an excel spreadsheet following the components of the outline and general publishing information. Salient points or paragraphs, with page references were pasted under appropriate outline areas. Almost 200 documents were reviewed and summarised using this template.

Table 10. Data extraction template

Search string		
Database		
Year		
Title /Citation		
Adopted by		
Abstract		
Internet link		
Peer reviewed		Y/N
Outline		
1	General	
2	Decisions for establishing stockpiles	2.1. Assessing the need for stockpiles – supporting national risk-assessment
		2.2 Identifying goals and objectives of potential stockpiles
3	Establishing stockpiles	3.1 Coordination and collaboration
		3.2 Stockpiles formulary
		3.3 Procurement strategies
		3.4 Legal
		3.5 Regulatory
		3.6. Storage facilities and infrastructure
		3.7. Financial considerations for set-up and maintenance
4	Operational planning	4.1. Stockpile maintenance
		4.2 Distribution and logistics
		4.3 Staff training and capacity building
5	Monitoring and evaluation	

A draft version of the document was initially developed and subsequently circulated among experts within WHO, as well as representatives from national and regional entities responsible for managing emergency stockpiles. An advanced draft was then made available for public consultation on the WHO website, during which additional feedback was solicited and systematically reviewed in accordance with WHO established procedures. Following this, the final draft was submitted for formal review through the appropriate WHO documentation review and approval processes.

Notes

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